

AUTOENCODER-BASED VISUAL REINFORCEMENT LEARNING IN CARRACING-V3

Raw-pixel PPO vs autoencoder-based latent PPO

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MOTIVATION AND PROBLEM STATEMENT

In Visual Reinforcement Learning, the agent receives images instead of compact state variables.



Raw pixel observations are high-dimensional



The agent must learn visual features and control at the same time

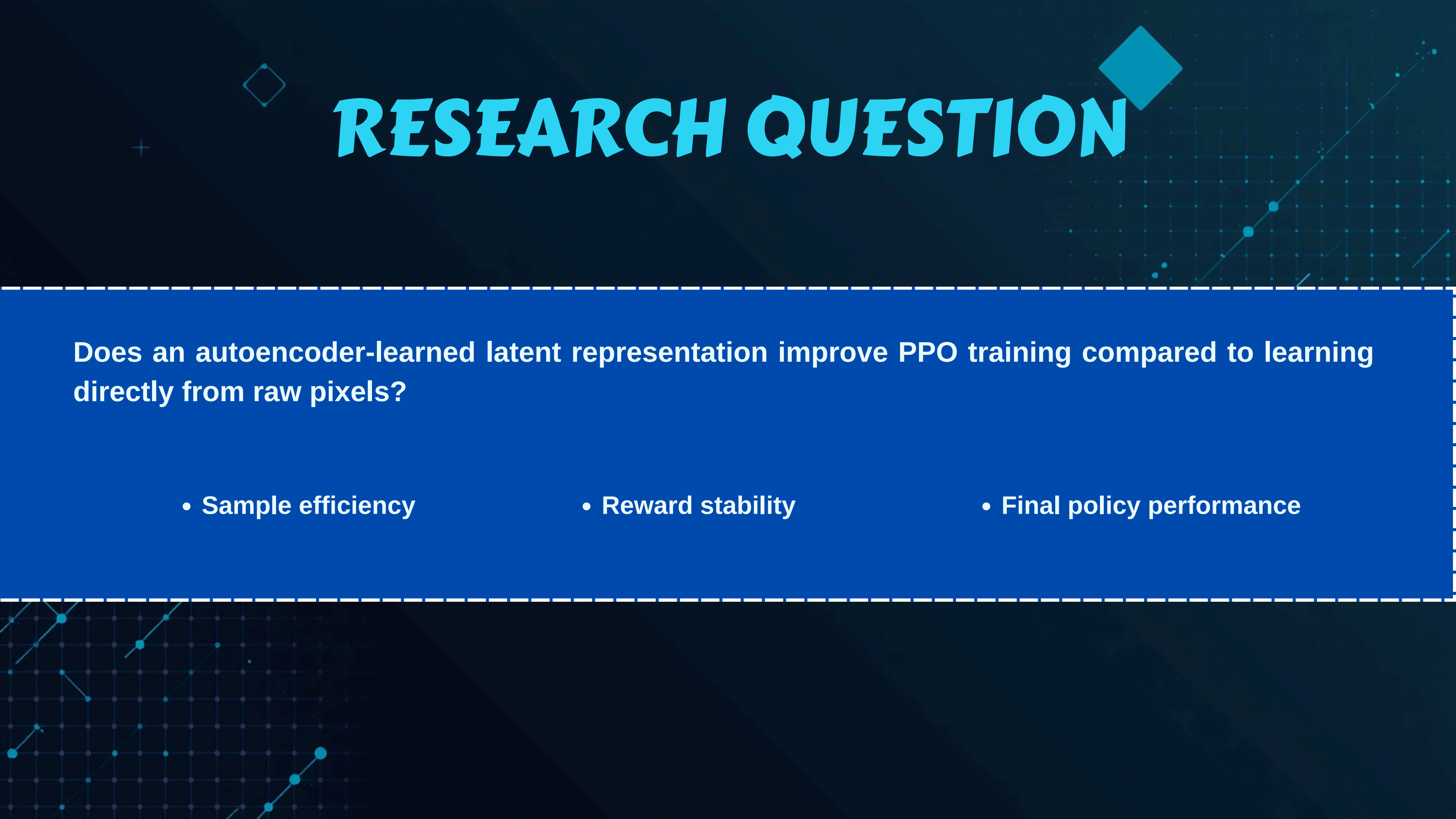


This can make PPO training unstable and sample inefficient



A learned latent representation may simplify the RL problem





RESEARCH QUESTION

Does an autoencoder-learned latent representation improve PPO training compared to learning directly from raw pixels?

- Sample efficiency
- Reward stability
- Final policy performance

ENVIRONMENT AND BASELINE

◆ CarRacing-v3

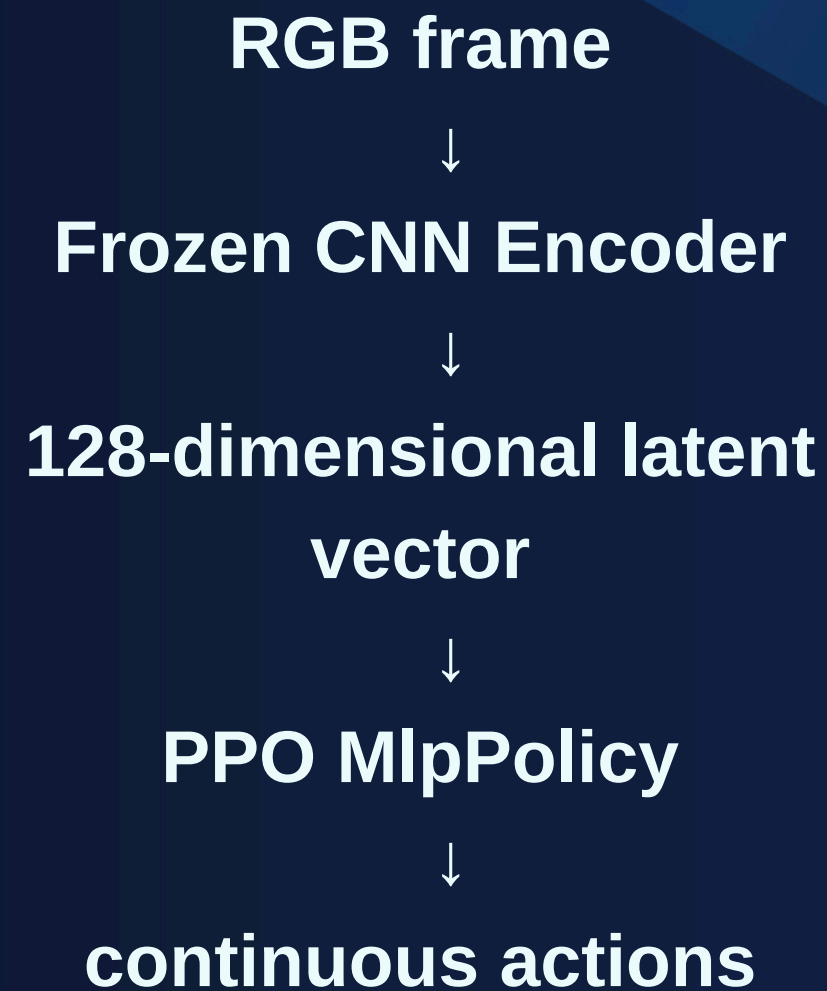
- Observation: RGB image, 96 x 96 x 3
- Action space: continuous
- Actions: steering, throttle, brake
- Episodes: up to 1000 steps

◆ Raw-Pixel PPO Baseline

- Stable-Baselines3 PPO
- CnnPolicy
- CNN feature extractor learned directly from reward
- Input: raw RGB frames

RGB frame → PPO CnnPolicy → action

PROPOSED METHOD: AUTOENCODER LATENT PPO



- ❖ Autoencoder training frames are sampled during PPO-environment interaction
- ❖ The autoencoder is trained before PPO
- ❖ Only the encoder is used during RL training
- ❖ The encoder weights are frozen
- ❖ PPO receives a compact latent state instead of pixels
- ❖ A Gymnasium wrapper converts each RGB observation into a latent vector

CNN AUTOENCODER

◆ Encoder

Input tensor: 3 x 96 x 96

- 32 x 48 x 48
- 64 x 24 x 24
- 128 x 12 x 12
- 256 x 6 x 6
- 128-dimensional latent vector

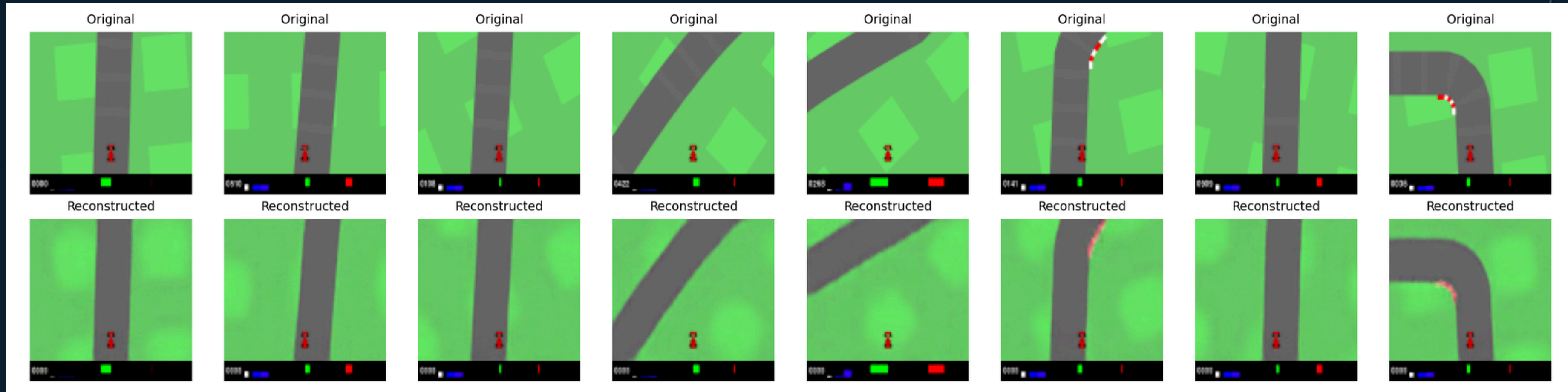
◆ Decoder

128-dimensional latent vector

- 256 x 6 x 6
- 128 x 12 x 12
- 64 x 24 x 24
- 32 x 48 x 48
- 3 x 96 x 96

Loss function: Mean Squared Error between original and reconstructed frame

AUTOENCODER RECONSTRUCTION RESULTS



- Road geometry is preserved
- Car position is preserved
- Fine details are slightly blurred

Blurring is expected with pixel-wise MSE reconstruction loss.

EXPERIMENTAL SETUP

Compared agents

- Raw-pixel PPO CnnPolicy directly from RGB frames
- Latent PPO Frozen encoder + MlpPolicy from 128-dimensional latent vectors

Training protocol

- Up to 2M timesteps per full run
- Evaluation every 10k timesteps
 - 5 deterministic evaluation episodes
- Continuous CarRacing-v3 action space

Metrics

- Best mean evaluation reward
- Final mean evaluation reward
 - Reward stability
 - Learning curve behavior

Data collection

- Frames saved during raw PPO training (~18k frames)
- Saved frames used as autoencoder dataset
- Same preprocessing used in the latent wrapper

Learning rates tested: $1e-4$ and $3e-4$

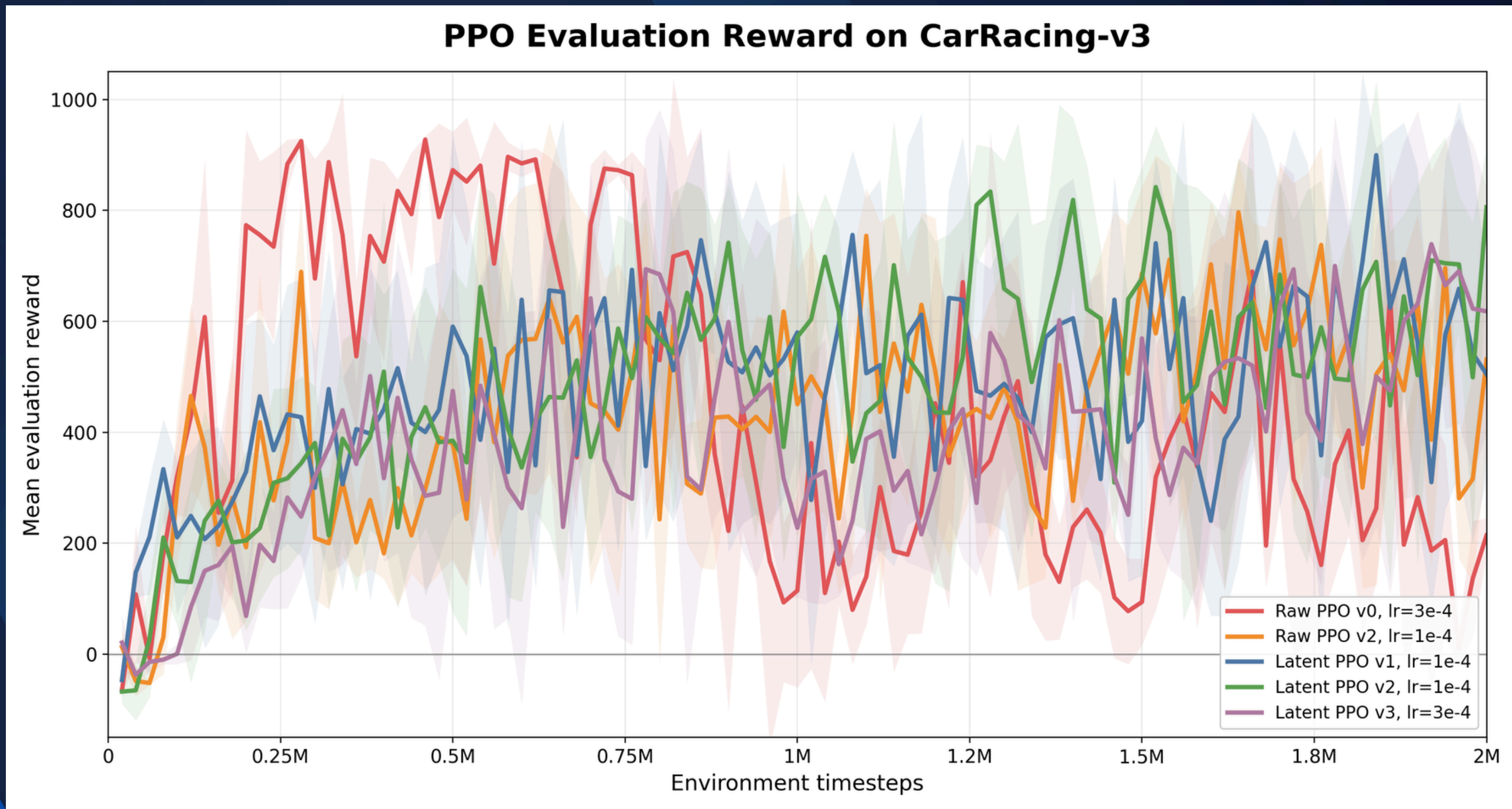
MAIN RESULTS

Run	Input	Learning rate	Best eval reward	Final eval reward
raw/v0	RGB pixels	3e-4	<u>927.78</u>	213.84
raw/v2	RGB pixels	1e-4	796.82	531.41
latent/v1	latent 128	1e-4	899.65	504.64
latent/v2	latent 128	1e-4	842.06	<u>805.81</u>
latent/v3	latent 128	3e-4	739.18	617.91

Highest peak reward: raw/v0

Best final reward: latent/v2

EVALUATION CURVES



Raw PPO reaches high peaks, while latent PPO is competitive but not consistently more sample-efficient.

INTERPRETATION

◈ Raw-pixel PPO

- Reached the highest peak reward
- Learned directly from visual input
- Showed strong instability in some runs

◈ Latent PPO

- Achieved competitive performance
- latent/v2 obtained the best final reward
- Did not consistently improve sample efficiency

Autoencoder latents are useful and compact, but reconstruction-based representations are not automatically better for control.

LIMITATIONS AND FUTURE WORK

Limitations

- Limited number of runs and random seeds
- CarRacing frames include HUD information
- MSE reconstruction produces slightly blurred images
- The encoder was frozen during PPO training
- No robustness tests in the final comparison

Future work

- Remove HUD using a crop wrapper
- Test robustness to noise, blur, and brightness changes
- Fine-tune the encoder during PPO training
- Compare different latent dimensions
- Use representation losses more aligned with control



THANK YOU